

# Summary Data on Acid Cropland and Lime Requirements in Nigeria

GAIA

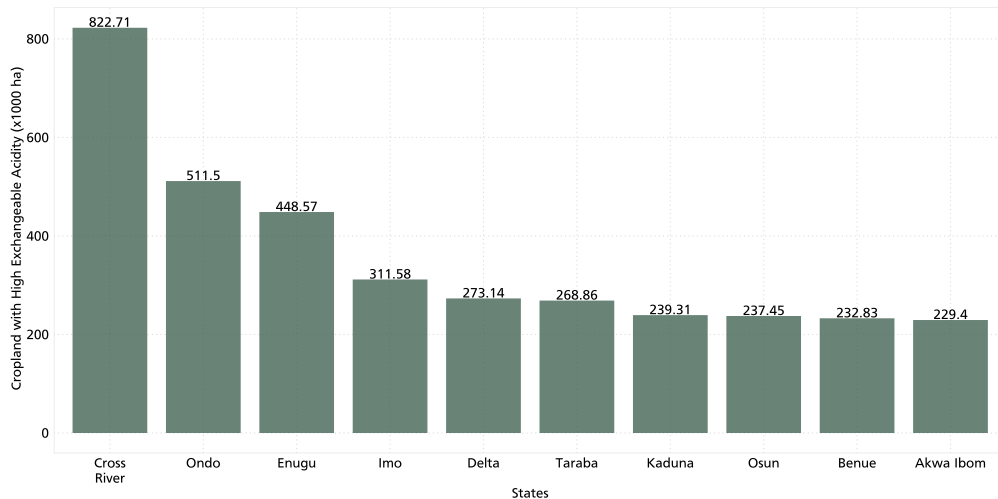
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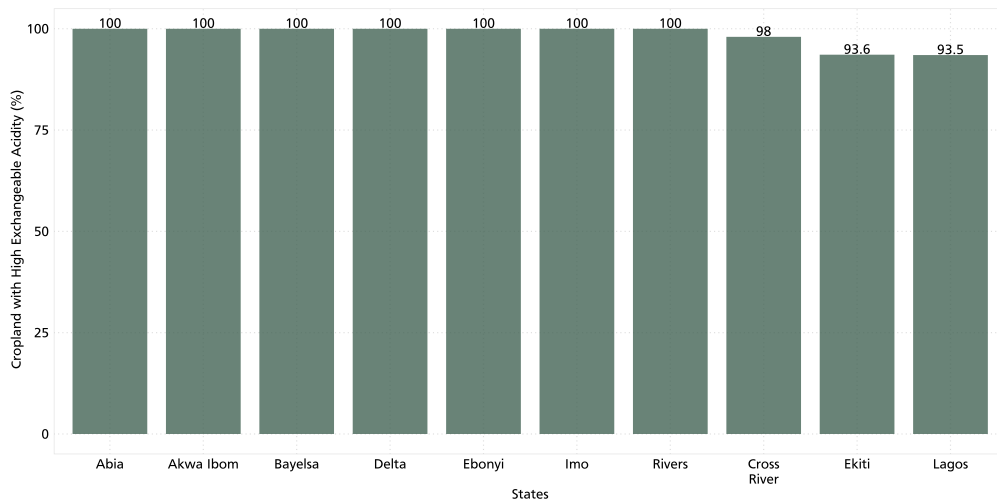
## Outlook

- 1 Introduction
- 2 Acidic Cropland
- 3 Yield Loss
- 4 Lime Requirement
- 5 Yield Response
- 6 Additional Production
- 7 ROI for Lime Application
- 8 Profitability of Lime Application

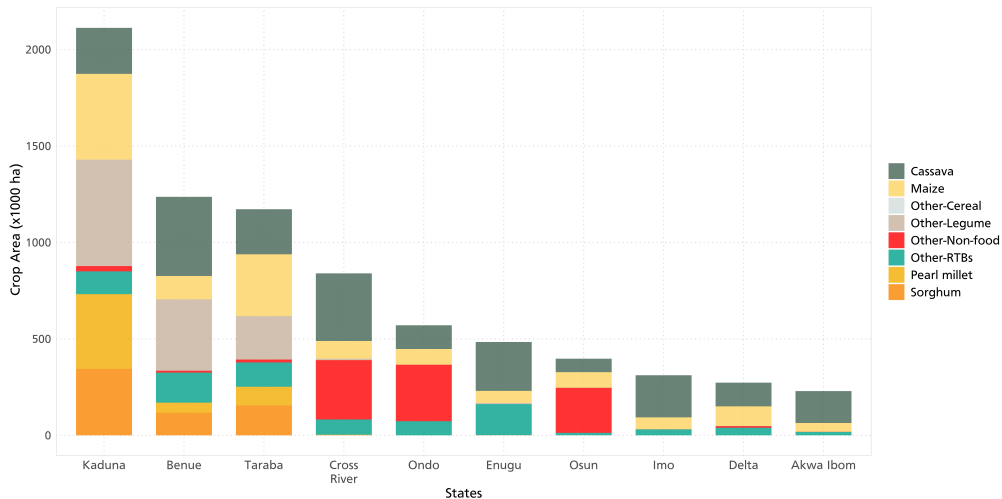
## Top 10 States in Nigeria with the highest acidic cropland



## Top 10 States in Nigeria with the highest acidic cropland (Percentage of total cropland)



## Total crop area in the top-10 most acidic States, not segregated by acidic cropland



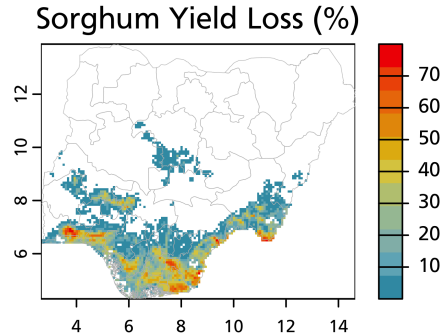
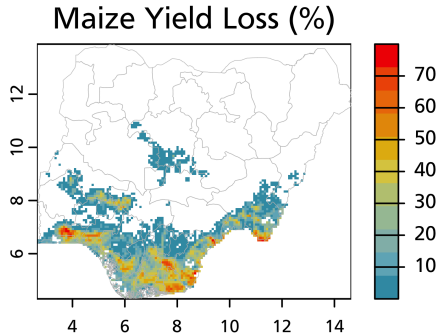
## Nigeria-Summary

└ Total crop area distribution (ha)

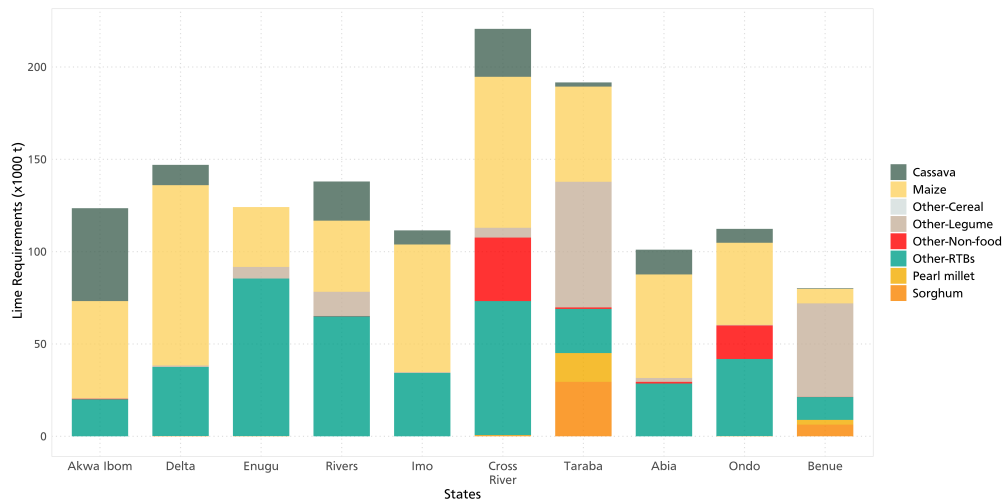
└ Total crop area in the top-10 most acidic States, not segregated by acidic cropland

| States      | Cassava | Maize  | Other-Cereal | Other-Legume | Other-Non-food | Other-RTBs | Pearl millet | Sorghum |
|-------------|---------|--------|--------------|--------------|----------------|------------|--------------|---------|
| Akwa Ibom   | 165.38  | 45.68  | -            | 0.16         | 0.80           | 17.38      | -            | -       |
| Benue       | 410.29  | 121.25 | -            | 369.94       | 9.80           | 156.01     | 52.17        | 117.09  |
| Cross River | 350.57  | 91.95  | -            | 7.53         | 307.53         | 79.02      | 0.96         | 2.24    |
| Delta       | 122.84  | 101.98 | -            | 0.65         | 8.22           | 39.21      | -            | 0.24    |
| Enugu       | 253.22  | 62.96  | -            | 5.92         | 0.03           | 160.64     | 0.27         | 0.77    |
| Imo         | 218.45  | 61.77  | -            | 0.31         | 0.47           | 30.58      | -            | -       |
| Kaduna      | 238.46  | 444.82 | 0.18         | 551.92       | 27.08          | 118.56     | 387.41       | 344.25  |
| Ondo        | 122.82  | 81.42  | -            | 0.29         | 292.60         | 73.25      | 0.06         | 0.16    |
| Osun        | 69.64   | 80.31  | -            | 1.04         | 233.53         | 12.15      | 0.04         | 0.50    |
| Taraba      | 233.59  | 320.93 | -            | 225.03       | 15.03          | 125.98     | 97.26        | 154.53  |

## Yield loss due to soil acidity for Maize and Sorghum in Nigeria



## Lime requirements (t) for acid soil remediation in Nigeria

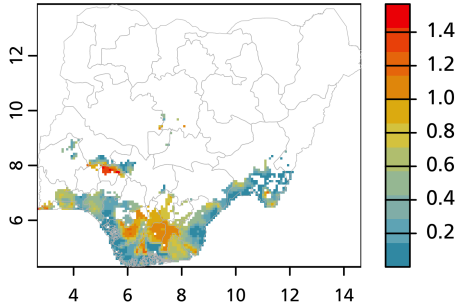


## Lime requirements (t) for acid soil remediation in Nigeria (table)

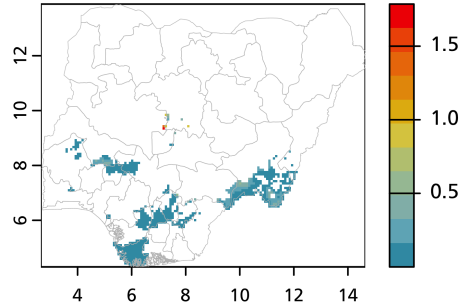
| States      | Cassava | Maize | Other-Legume | Other-Non-food | Other-RTBs | Sorghum | Pearl millet | Other-Cereal |
|-------------|---------|-------|--------------|----------------|------------|---------|--------------|--------------|
| Abia        | 13.42   | 56.06 | 2.12         | 0.90           | 28.55      | 0.02    | -            | -            |
| Akwa Ibom   | 50.30   | 52.71 | 0.19         | 0.28           | 20.05      | -       | -            | -            |
| Benue       | 0.33    | 7.88  | 50.63        | 0.08           | 12.37      | 6.35    | 2.56         | -            |
| Cross River | 25.98   | 81.76 | 5.30         | 34.36          | 72.56      | 0.45    | 0.24         | 0            |
| Delta       | 10.98   | 97.69 | 0.77         | 0.01           | 37.37      | 0.18    | -            | -            |
| Enugu       | -       | 32.43 | 6.28         | -              | 85.32      | 0.11    | 0.01         | -            |
| Imo         | 7.64    | 69.19 | 0.38         | 0.01           | 34.31      | 0.01    | -            | -            |
| Ondo        | 7.49    | 44.52 | 0.28         | 18.13          | 41.79      | 0.08    | 0.04         | -            |
| Rivers      | 21.22   | 38.50 | 13.10        | 0.15           | 65.02      | 0.01    | -            | -            |
| Taraba      | 2.27    | 51.52 | 68.05        | 0.80           | 23.90      | 29.50   | 15.61        | 0            |

## Lime rate for Maize and Sorghum in Nigeria

Maize Lime Rate (t/ha)

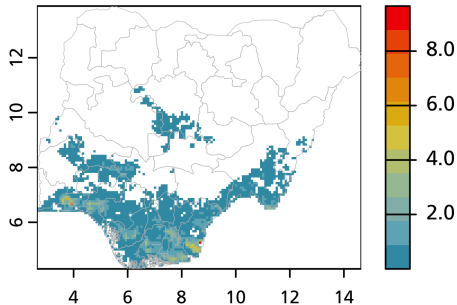


Sorghum Lime Rate (t/ha)

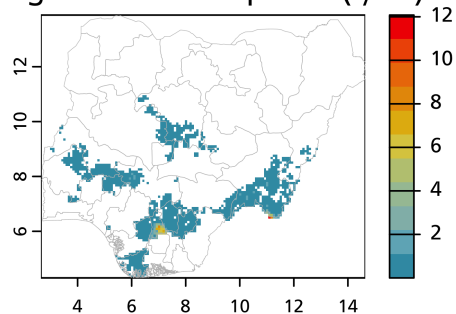


## Yield response to lime application in Nigeria

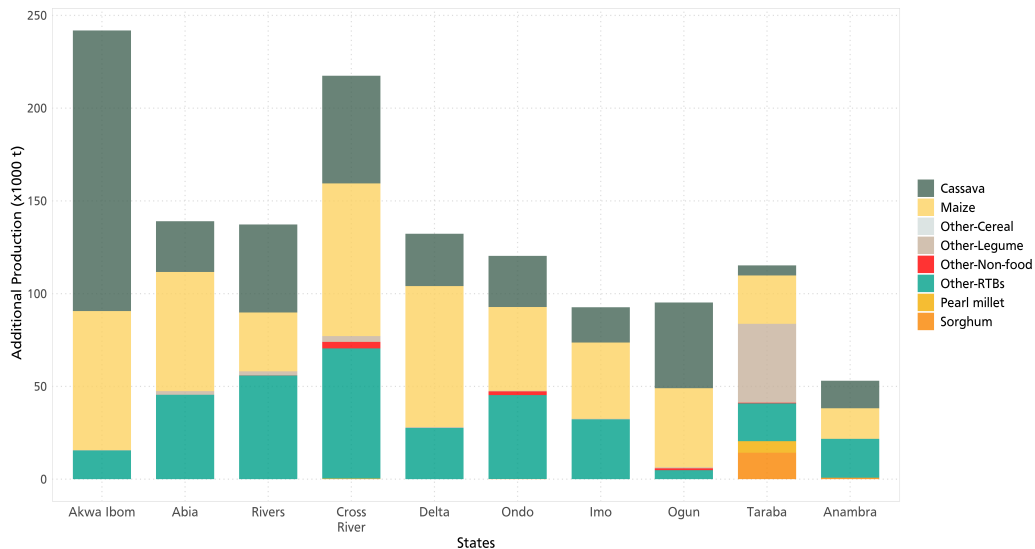
Maize Yield Response (t/ha)



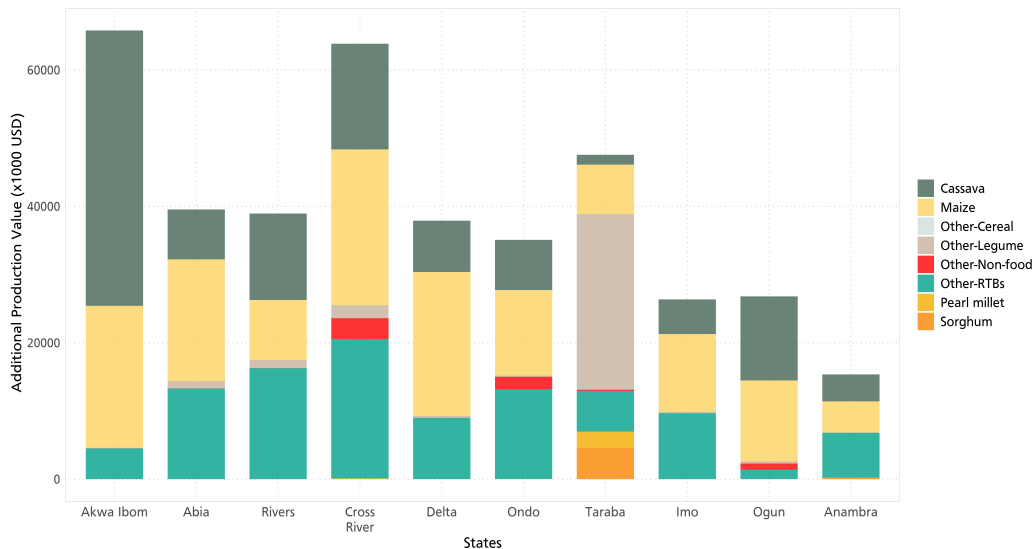
Sorghum Yield Response (t/ha)



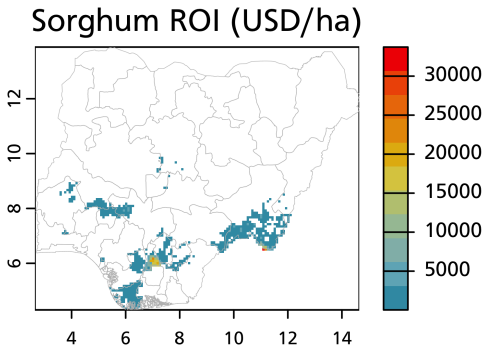
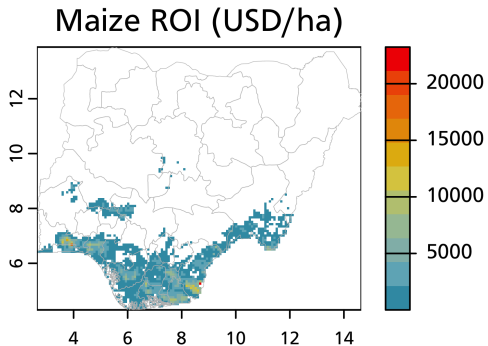
## Potential for additional production (tone) in top 10 acidic States in Nigeria



## Potential for additional production value (US\$) in top 10 acidic States in Nigeria

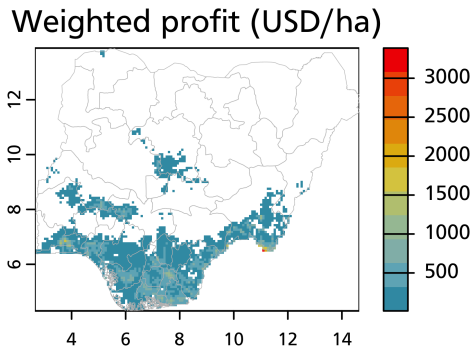
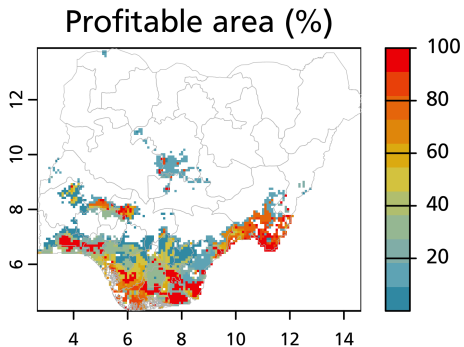


## Return on investment for lime application in Nigeria



- └ Profitability of Lime Application
- └ Profitability of lime application in Nigeria

## Profitability of lime application in Nigeria



## Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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